



Started on	Thursday, 20 August 2026, 8:54 AM
State	Finished
Completed on	Thursday, 20 August 2026, 8:56 AM
Time taken	1 min 43 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  void function(int n) {
4      int count = 0;
5
6      count++;
7      for (int i = n / 2; i < n; i++) {
8
9          count++;
10         for (int j = 1; j < n; j = 2 * j) {
11
12             count++;
13             for (int k = 1; k < n; k = k * 2) {
14                 count++;
15                 count++;
16             }
17             count++;
18         }
19         count++;
20     }
21
22     printf("%d\n", count+1);
23 }
24
25 int main() {
26     int n;
27     if (scanf("%d", &n) == 1 && n > 0) {
28         function(n);
29     }
30     return 0;
31 }
32
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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